

The Utility Problem

Utilities often have error-ridden connectivity data describing the interconnections of the components and assets of their distribution grid, which can lead to errors such as topological errors or islanded assets (separated from all sources). Any analysis of the grid with this “dirty” data can lead to incorrect conclusions, therefore rendering the analysis useless and potentially detrimental. In order to correct this data, data employees are often tasked with manually sorting through connectivity data to identify errors and correct them.

This data problem represents two major issues: the labor and time required from data professionals to manually sift through data as well as the inability of the organization to gather valuable insights from their grid data. Not only is the labor of a data professional a significant cost on the utility, but it also takes away an experienced skillset that could be used to tackle more complex data challenges. In addition, this manual process results in a much longer timeline to correct the data. Without “clean” data, the utility cannot perform insightful analysis, thus hampering their abilities to manage countless data-driven challenges such as managing the energy transition and improving grid reliability.

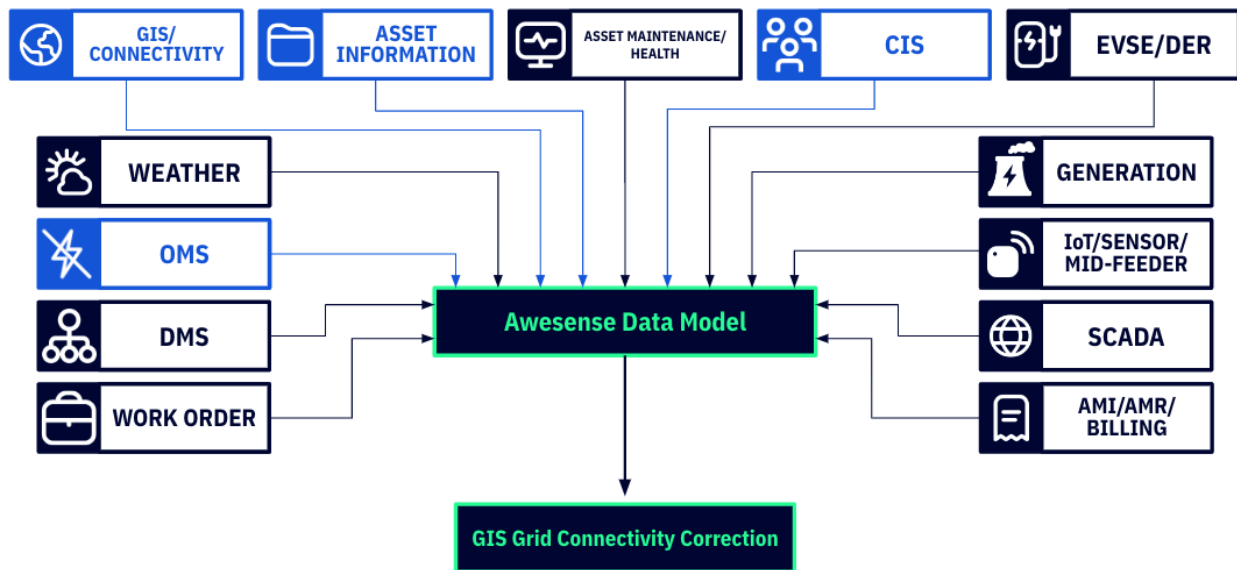
The Data Solutions and Methods

An algorithm can identify and correct data connectivity errors by first identifying islanded assets and then reconnecting them to the proper part of the grid. These corrections can then be confirmed by reviewing the connectivity model through sampling by data professionals.

The Awesense AI Data Engine solution comprises a Validation, Estimation, and Error Correction - (VEE) engine. This algorithm sifts through the data: “validating” the correctness of the connectivity data; “estimating” fixes to the errors identified and “correcting” the remaining errors to provide a more accurate depiction of the grid. This all is supported at scale. GIS/Connectivity, OMS and CIS data are ingested into the data engine and can then be corrected through parameterization. The parameters can be set by the user in order to determine the intensiveness of the algorithm. This algorithm is able to identify and correct connectivity gaps, islanded assets, incorrect sources and other common connectivity issues. The algorithm can be deployed multiple times to increase the accuracy of the model where the output of the validation and correction can be fed back into the source system for further improvement.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. This use case is implemented as part of the Awesense AI Data Engine. Please contact [Awesense](#) if you would like to see a demo of this use case.